

Written Assignment 2

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Problem #1

4 servers - random day in April (30 days)

$$1.a \quad P(E) = \frac{1}{30} \times \frac{1}{30} \times \frac{1}{30} \times \frac{1}{30} = \frac{1}{21,000} \text{ same day}$$

s1 s2 s3 s4

$$1.b \quad P(\text{distinct day}) = \frac{30 \times 29 \times 28 \times 27}{30^4} \approx 0.812 \text{ distinct day}$$

$$1.c \quad P(\text{2 on same day}) = 1 - p$$
$$1 - 0.812 = 0.188 \text{ same day}$$

Problem #2

Average rate λ of 20 request per second

Poisson distribution

↳ probability of specific # of events

$$f(x) = \frac{\lambda^x e^{-\lambda}}{x!}$$

λ = expected # of events

$e = 2.71828$

x = number of events

2.a $x = 1000$

$\lambda = 20$ request a second
60 sec in a min

$$20 \times 60 = 1200$$

$$f(1000) = \frac{1200^{1000} e^{-1200}}{1000!} = \text{overflow}$$

2.b $\Pr[\emptyset \text{ arrivals in 3 sec}]$

$$\lambda = 20 \times 3 = 60$$

$$P(0) = \frac{60^0 e^{-60}}{0!}$$

$$\approx 9.7565 \times 10^{-27}$$

2.c $x = 20$ sec

$$\lambda = 20 \times 0.5 = 10$$

$$P(1) = \frac{10^{20} e^{-10}}{20!} \approx 0.001866$$

2.d 10% of the time busy on average
average service rate = μ

$$p = \frac{\lambda}{\mu}$$

$$p = 0.10$$

$$\lambda = 20 \text{ sec}$$

$$\frac{20}{0.10}$$

= 20.57 average
service rate

Problem #3

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Pseudo random number generator

3.a integers b/t 0 and 99

```
int rand_0_to_99() {  
    return rand() % 100;  
}
```

3.b floating integers b/t 0.25 + 0.5

```
float rand_0.25_0.5() {
```

```
    return 0.25 + ((float)rand() / RAND_MAX) * 0.25;  
}
```

Shift by 0.25
instead of 0

b/t 0 + 1

0.5 - 0.25

3.c 1 = prob 0.5
2 = prob 0.2

float = b/t 3 + 4

0.5 - 0.7 = 0.2 (20%)

```
float rand_conditions() {  
    float r = ((float)rand() / RAND_MAX);
```

```
    if (r < 0.5) {  
        return 1;
```

```
    } else if (r < 0.7) {  
        return 2;
```

```
    } else {  
        return 3 + ((float)rand() / RAND_MAX);  
    }  
}
```